Smart Street Light Control Network

Capstone Review 1 | Implementation & Architecture Overview

S Hariram
Reg No: 1965260018

Guru Santhosh S
Reg No: 1924260072

Nitin C
Reg No: 1965260065

Course
Computer Networks

Course Code
CSA0703

Faculty Advisor
Dr. K. Jaisharma

Modernizing Municipal Infrastructure

A municipal corporation requires a robust system for street lights to switch on and off automatically and be centrally monitored. The goal is to drastically save electricity across the locality.

IoT Integration

Simulated controller nodes for intelligent operations.

Centralized Control

Automated scheduling via Python client-server architecture.

Data-Driven Decision Making

Verification of energy savings using network traffic analysis.

Problem Identification & Analysis

Manual Operations

Reliance on manual labor or static timers leads to lights remaining active unnecessarily during bright daylight hours.

Lack of Monitoring

Inability to detect lighting faults or measure exact uptime performance across specific geographic sectors.

No Centralization

Fragmented local control makes locality-wide municipal policy changes slow, cumbersome, and expensive.

Solution Design Overview

1. Network Design

Simulating 5 street-light controller nodes connected to a central control server to guarantee reliable network reachability.

2. Control Program

Developing a custom Python client-server application to broadcast scheduled ON/OFF commands to localized nodes.

3. Verification

Capturing command traffic using Wireshark to calculate off-durations and generate comprehensive energy-saving reports.

STUDENT 1

Implementation: Module 1

Street Light Node Network Design & Topology Setup

Street Light Node Network Design

Establishing the core infrastructure required for centralized municipal communications.

Tasks

Python code with 5 simulated street-light controller nodes connected to a central server.

Techniques

Packet Tracer environment, IoT node simulation and subnet mapping.

Outcome

A reliable network topology where all simulated light nodes remain accessible via a structured IP addressing scheme.

```
] Listening on 127.0.0.1:5050
] Waiting for 5 nodes to connect...
1] Connected and registered.
] Registered NODE-01 from ('127.0.0.1', 52556)
2] Connected and registered.
] Registered NODE-02 from ('127.0.0.1', 52557)
3] Connected and registered.
] Registered NODE-03 from ('127.0.0.1', 52558)
4] Connected and registered.
] Registered NODE-04 from ('127.0.0.1', 52559)
5] Connected and registered.
] Registered NODE-05 from ('127.0.0.1', 52560)
] All 5 nodes connected. Starting schedule.
```

STUDENT 2

Implementation: Module 2

Centralized ON/OFF Control Program Development

Centralized ON/OFF Control Program

The software backbone for intelligent switching based on dynamic time schedules.

Tasks

Write a Python client-server program sending automated ON/OFF commands based on preset time schedules.

Techniques

Python sockets, robust network loops, and scheduled command broadcasting.

Outcome

5 target nodes correctly switch state aligning with server-delivered triggers, achieving high packet delivery success.

```
1] Hour 06:00 -> state changed to OFF
2] Hour 06:00 -> state changed to OFF
3] Hour 06:00 -> state changed to OFF
4] Hour 06:00 -> state changed to OFF
5] Hour 06:00 -> state changed to OFF
] Hour 06:00 -> broadcast 'OFF' to 5 nod
1] Hour 18:00 -> state changed to ON
2] Hour 18:00 -> state changed to ON
3] Hour 18:00 -> state changed to ON
4] Hour 18:00 -> state changed to ON
5] Hour 18:00 -> state changed to ON
] Hour 18:00 -> broadcast 'ON' to 5 node
```

STUDENT 3

Implementation: Module 3

Energy-Saving Verification Report & Traffic Analytics

Energy-Saving Verification Report

Quantifying the operational and environmental impact of the newly deployed smart control systems.

Tasks

Capture active traffic in Wireshark and calculate estimated 'off' durations per node across the grid.

Techniques

Wireshark packet capture analysis, statistical uptime calculation, and reporting metrics.

Outcome

A comprehensive report estimating the percentage of electricity saved via centralized scheduling.

```
=====================================================
SAVING VERIFICATION REPORT
treet Light Control Network — Capstone Review 1
=====================================================

     OFF Hours / 24h      Est. Energy Saved
---------------------------------------------
     12                   50.0%
     12                   50.0%
     12                   50.0%
     12                   50.0%
     12                   50.0%
---------------------------------------------

-wide average energy saved:  50.0%
 delivery success rate:      100.0%
ommands captured:             10
```

Project Parameter Summary

Module Component	Specific Parameters	Expected Outcome
Network Design (Student 1)	No. of nodes (5), link type, IP addressing scheme	Full node reachability from central server
Control Program (Student 2)	Schedule trigger times, delivery success rate %	Lights correctly switch state on schedule
Verification Report (Student 3)	Scheduled off-hours, total lights, estimated saving %	Data-backed report demonstrating energy benefit

End-to-End Architectural Workflow

01

Topology Build

Deploying IoT nodes and verifying physical/logical connections within Packet Tracer simulator.

02

Socket Execution

Running the server backend to broadcast automated timing schedules directly to client controller nodes.

03

Impact Metric

Validating savings and confirming packet reliability using Wireshark capture logs and data analysis.

Project Conclusion & Future Scope

- **Proven Feasibility:** Successfully demonstrated automated municipal street lighting using lightweight socket communication and topology simulation.
- **Measurable Savings:** Estimated reduction in power consumption by eliminating daylight wastage across simulated nodes.
- **Future Scalability:** Expansion path towards cloud-based MQTT protocols, light-sensor feedback loops, and automated fault reporting.

Thank You!

Ready for questions and feedback from the review panel.

Smart Street Light Control Network | Capstone Review 1